

Decentralized Observation: Unanimity, Deadlock, and the Value of Communication

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Abstract

Institutions observe different indicators and command different instruments; the obstruction calculus has so far spoken with one regulator's voice. This paper, the D13 item of the generalization catalogue, models two agencies with split observation rights (agency 1 reads z_1 , agency 2 reads z_2) and a unanimity command protocol: the instrument executes only on agreement, and disagreement deadlocks the system. On the two-patch working system: the coordinator (reading the joint state) sustains 28 of 36 safe pairs; the decentralized unanimity regime, enumerated exhaustively over all 256 agency-law pairs, sustains exactly 12 — and exactly 16 coordinated-viable pairs are lost, including an exact instance viable under each agency alone and under the coordinator but nonviable under *every* unanimity law pair. The audited belief $\{(1, 2), (2, 1)\}$ survives decentralization only through an anti-symmetric voting law, and the authority restriction (forbidding agency 2's vote) destroys the entire decentralized kernel ($12 \rightarrow 0$, strict instance: the audited belief). The obstruction is robust to one-step delayed common knowledge. Communication has exact, computable value: the deadlock shadow (6/6 check families).

Introduction and relation to the programme

The catalogue proposes multiple regulators with split observation rights $y_i = O_i(x)$ and split command rights $u_i \in U_i$, joint action constrained by common knowledge, delays, authority, and enforcement — and warns that it is a larger extension to be added only where institutional constraints are central. The governance focus of the programme is exactly that case; the paper stays minimal — two agencies, one instrument, unanimity — and computes exactly.

Split rights under unanimity

The realized instrument is the agreed vote: $u = u_1(z_1)$ when $u_1(z_1) = u_2(z_2)$, deadlock ($u = 0$, both patches damaged) otherwise. The induced closed loop is deterministic for a fixed law pair (u_1, u_2) , so a belief is decentralizable

iff some law pair's trajectory never exits the safe set (cycle detection on belief sets, exact).

Proposition 1 (The deadlock obstruction and its exact boundary). The coordinator's kernel holds 28 of 36 safe pairs; the decentralized kernel holds exactly 12; exactly 16 coordinated-viable pairs are not decentralizable, and one such instance is exhibited that is viable under each agency alone (the z_1 -only and z_2 -only structures of the successor) and under the coordinator. The audited belief $\{(1, 2), (2, 1)\}$ is decentralizable only through voting laws including the anti-symmetric one (16 of 256 law pairs work) — institutional coordination is load-bearing exactly on the deadlock shadow.

Proof. Exhaustive enumeration over the 256 law pairs with exact trajectory simulation; the exhibited instance and the anti-symmetric law's trajectory are verified directly. The counting identity $28 - 12 = 16$ is the communication value on this family. $\square$

Proposition 2 (Authority and delay). Removing agency 2's right to vote 2 collapses the decentralized kernel from 12 to 0 (strict instance: the audited belief), and granting agency 2 a one-step delayed reading does not restore the deadlock boundary: the decentralized kernel under delayed common knowledge remains strictly below the coordinator's.

Proof. Both computations are exhaustive and exact: the restricted law space is the 16 pairs with $u_2 \equiv 1$; the delayed regime simulates the augmented (state, previous- z_2) trajectories. $\square$

Verified record (6/6 exact check families)

E1 the coordinator's kernel (28, the successor's full-observation count); E2 the decentralized kernel (12) by exhaustion over 256 law pairs; E3 the exact boundary (16 lost pairs), the exhibited individually-sufficient-yet-jointly-nonviable instance, and the audited belief's anti-symmetric survival; E4 the setwise communication value ($12 \subset 28$); E5 the authority collapse ($12 \rightarrow 0$, strict instance recorded); E6 the one-step-delay robustness of the deadlock boundary.

What is not claimed

Two agencies, one instrument, unanimity: no n -agency theory, no sequential or weighted voting, no mechanism design, no enforcement or compliance dynamics, no negotiation models. The deadlock protocol is the simplest common-knowledge discipline; procedural alternatives (default rules, tie-breakers) are not classified. Institutional effectiveness is not estimated from data. The catalogue’s larger-programme warning is adopted: the paper claims the exact deadlock phenomenon and its boundary on the audited system, nothing broader.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `institutional_observation_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.